

# Howard Wei

[howardwei21@gmail.com](mailto:howardwei21@gmail.com) | [hwei.vercel.app](https://hwei.vercel.app) | [linkedin.com/in/weihoward](https://linkedin.com/in/weihoward) | [github.com/HowardWei123](https://github.com/HowardWei123) | San Jose, CA

## EDUCATION

### San Jose State University

San Jose, CA

*B.S. in Computer Science, Minor in Business*

*Expected Graduation: May 2027*

**Relevant Coursework:** Artificial Intelligence, Machine Learning, Processing Big Data, Information Security

## EXPERIENCE

### Cytokinetics

South San Francisco, CA

*Clinical Systems Specialist*

*Sept 2025 – Present*

- Engineering a **HIPAA-compliant** multi-agent architecture via **LangChain** for clinical data system risk analysis.
- Streamlined clinical data review using **Python** and **Pandas**, cutting outstanding review time by **10 days**.

### Cytokinetics

South San Francisco, CA

*Informatics Intern*

*May 2025 – Aug 2025*

- Developed an end-to-end agentic AI data dashboard application with **AWS** (Bedrock, Lambda, S3) and **Python**, used by hundreds of company scientists to create multiple visualizations at once for millions of data points.
- Created a **Spotfire** IronPython-JavaScript API system to generate advanced scientific visualizations for complex chemical structures, reducing application load times by **50%**.
- Integrated a **PostgreSQL**-backed memory layer to preserve user-agent context across concurrent sessions.

### Google Developer Group

Merced, CA

*Software Engineering Lead Developer*

*Sept 2024 – Dec 2024*

- Deployed a paper trading platform for analyzing trading habits and providing retail investors with insights through **AWS** (EC2, RDS, Lambda).
- Recruited and directed a **25-member** ML team using **Agile** methodologies (Kanban) and **Jira**, owning **CI/CD** pipelines that automated model testing and deployment.
- Implemented a data **ETL** pipeline to Llama 3 70B API through **Django REST** and **MySQL** market data sources, improving financial advice accuracy by **35%** across 7 different metrics.

### Machine Learning at SJSU

San Jose, CA

*Software Engineer*

*Sept 2024 – May 2025*

- Developed **TypeScript** API endpoints for club website to store and retrieve user data from **PostgreSQL**.
- Built frontend with **React** and **Tailwind CSS**, allowing member content sharing and event publishing.

## PROJECTS

### F1TelemetryML – Python, GCP, PyTorch, XGBoost, FastF1, Pandas, NumPy, Matplotlib

- Orchestrated an **end-to-end Python pipeline** using FastF1 to process **146,000+** corner telemetry instances across 10 Grand Prix tracks to train Ridge, Random Forest, MLP, and LSTM models on raw sensor inputs.
- Predicted **3-axis driver-style scores** across in-distribution and unseen track conditions, achieving a top test **MAE** of **0.0259** with an LSTM architecture.

### SpotlightAI – Python, GCP, ChromaDB, FastAPI, Postman, GeoPy, PostgreSQL, Streamlit

- Organized development of an **end-to-end system architecture** of a review-based restaurant recommendation RAG model with additional location-based capabilities.
- Integrated **ChromaDB** with recent Google Places API data and historical Yelp reviews to enable **retrieval-augmented generation** for Gemini, improving response accuracy and reducing hallucinations.

### EmotionAI – Python, PyTorch, OpenCV, Pandas, Matplotlib, Streamlit

- Led 5 student engineers across 4 universities in developing a Python-OpenCV human emotion detection model.
- Deployed on **Streamlit** with user-AI interactions through webcam, achieving **60%+** accuracy in live testing.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, Java, C, C++, MATLAB, R, SQL, Bash

**Frameworks:** LangChain, CrewAI, PyTorch, Streamlit, Django, React, Vue, Tailwind CSS

**Developer Tools:** AWS, GCP, Azure, Git, Jira, Docker, Firebase, Postman

**Libraries:** ChromaDB, OpenCV, Keras, Matplotlib, Seaborn, NumPy, Pandas, Scikit, Pyodide, SWR, XGBoost

**Interests:** Bioinformatics, Data Analytics, Formula 1, LLM Inference, Rock Climbing, Systems Architecture